ProjFlow: Projection Sampling with Flow Matching for Zero-Shot Exact Spatial Motion Control

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Abstract

*Precise spatial control in human motion generation—such as following joint trajectories or satisfying keyframe constraints—currently requires task-specific retraining or slow inference-time optimization that often fails to satisfy constraints exactly. We present **ProjFlow**, a training-free projection sampler for flow-matching models that achieves zero-shot exact spatial motion control. Our key insight is that many practical motion control tasks can be formulated as linear inverse problems, enabling closed-form data-consistent updates during sampling. ProjFlow enforces constraints by projecting each flow step's clean motion estimate onto the feasible set using a novel kinematics-aware metric that spreads corrections coherently along the skeleton topology. We introduce time-varying pseudo-observations with dynamic masking and adaptive variance to handle sparse constraints like long-gap inpainting. On trajectory control tasks, ProjFlow achieves perfect constraint satisfaction (0.0000 error) while maintaining competitive realism (FID 0.107 for pelvis control, 0.097 for all joints), significantly outperforming the prior zero-shot method DNO (FID 0.151) and matching trained controllers' quality without retraining. For 2D-to-3D lifting, we achieve exact 2D reprojection (MPJPE-2D = 0.000) with strong 3D plausibility (MPJPE-3D 0.042), improving over existing approaches. ProjFlow demonstrates that precise, zero-shot spatial control is achievable through principled projection sampling with kinematics-aware metrics.*

1. Introduction

Human motion generation has achieved remarkable progress in producing realistic and diverse movements from textual descriptions [3, 10]. However, many practical applications require *precise spatial control*—for instance, ensuring that a character follows a specific hand trajectory when "drawing a heart" or satisfying exact keyframe constraints during motion editing. Current approaches face a fundamental tradeoff: methods that guarantee exact constraint satisfaction typically require task-specific retraining [11], while zero-shot approaches sacrifice precision for generality, often leaving residual constraint violations [5].

This limitation stems from the underlying tension between preserving the learned motion manifold and enforcing hard constraints. Training-based controllers can learn to jointly optimize both objectives but lack flexibility and require extensive retraining for new constraint types. Inference-time optimization methods [5] provide more flexibility but rely on iterative solvers that converge slowly and may not reach exact feasibility, particularly for high-dimensional constraint spaces.

We propose **ProjFlow**, a training-free projection sampler that achieves *zero-shot exact spatial control* for flow-matching motion models. Our approach is grounded in the observation that many practical motion control tasks can be formulated as linear inverse problems of the form $\mathbf{y} = \mathbf{A}\mathbf{x} + \boldsymbol{\epsilon}$, where $\mathbf{A}$ represents spatial measurement operators (e.g., joint selection, camera projection) and $\mathbf{y}$ contains the target observations.

The core insight is to enforce constraints via closed-form projections during flow sampling. At each sampling step, we predict the clean endpoint motion, then apply a data-consistent correction that exactly satisfies the constraints while minimizing a carefully designed regularization norm. Crucially, we introduce a *kinematics-aware metric* based on skeletal topology that distributes constraint corrections coherently across the kinematic chain, preserving motion naturalness.

For sparse constraints such as motion inpainting, we develop a time-varying pseudo-observation scheme with dynamic masking and adaptive variance. This allows the sampler to smoothly interpolate through missing regions while respecting the available anchor points with adjustable precision.

Our contributions are:

- A unified framework for spatial motion control as linear inverse problems solved via projection-based flow sampling.
- A kinematics-aware regularization metric that respects skeletal topology for natural constraint handling.
- Time-varying pseudo-observations with dynamic masking for robust sparse constraint handling.
- Zero-shot exact constraint satisfaction without retraining,

achieving competitive motion realism.

We demonstrate ProjFlow on trajectory control and 2D-to-3D motion lifting tasks. On HumanML3D [3], ProjFlow achieves perfect constraint satisfaction (trajectory/location/average errors of 0.0000) while maintaining strong realism (FID 0.097–0.107), significantly outperforming the zero-shot baseline DNO [5] (FID 0.151) and matching the quality of trained approaches without their inflexibility.

2. Related Work

Text-to-Motion Generation. Recent advances in diffusion models have enabled high-quality human motion synthesis from text descriptions. MDM [10] pioneered the application of denoising diffusion to motion generation, while subsequent works like PriorMDM [9] and GMD [4] improved quality and controllability. Large-scale datasets like HumanML3D [3] have been crucial for training these models. Recent works have also explored flow matching as an alternative to diffusion [7].

Spatial Motion Control. Controlling the spatial aspects of generated motion remains challenging. Training-based approaches like OmniControl [11] and MaskControl [8] learn to condition generation on spatial constraints but require retraining for new control signals and may not guarantee exact satisfaction. ControlNet-based approaches [2] adapt image generation control methods but suffer from similar limitations.

Inference-Time Optimization. Zero-shot control methods avoid retraining by optimizing objectives during inference. DNO [5] uses a differentiable neural optimizer to iteratively adjust motion predictions, while other approaches employ gradient-based optimization [4]. However, these methods typically converge slowly and may not achieve exact constraint satisfaction.

Projection Methods for Generative Models. Projection-based sampling has been explored in various generative modeling contexts. DPS [1] projects intermediate samples in diffusion models for image inverse problems. DDRM [6] focuses on linear degradation operators with efficient SVD-based projections. ADMM-inspired approaches [**?**] employ alternating optimization for non-linear constraints but require multiple forward passes per step.

Recent flow matching extensions include FlowPG [**?**] for physics-guided generation and ConstrainedFM [**?**] for optimization-free constraint handling in continuous domains. However, these methods either lack exact constraint guarantees or require domain-specific architectural modifications. Our work provides the first unified framework for exact linear constraint satisfaction in discrete motion generation through kinematics-aware projection sampling.

Physics-Based Motion Control. Physics-based approaches [12] incorporate biomechanical priors through differential simulation but struggle with computational efficiency and constraint flexibility. Recent neural physics methods [**?**] achieve impressive realism but require extensive physics tuning and cannot easily adapt to novel constraint types, highlighting the value of our geometry-agnostic projection framework.

3. Preliminaries

Flow Matching for Motion Generation. We build upon flow matching models for text-conditioned motion generation. Given a text prompt $\mathbf{c}$ and target motion $\mathbf{x}_1 \in \mathbb{R}^{D\times L\times J}$ (with D spatial dimensions, L frames, and J joints), flow matching learns a time-dependent velocity field $\mathbf{v}_\theta(\mathbf{x}_t, t, \mathbf{c})$ that transforms noise $\mathbf{x}_0 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ to data via the ODE:

$$\frac{d\mathbf{x}_t}{dt} = \mathbf{v}_\theta(\mathbf{x}_t, t, \mathbf{c}), \quad t \in [0, 1] \tag{1}$$

For the linear rectified flow path with $\alpha_t = t$ and $\sigma_t = 1 - t$, the velocity field prediction allows recovery of clean endpoint estimates:

$$\hat{\mathbf{x}}_1 = \mathbf{x}_t + (1 - t)\mathbf{v}_\theta(\mathbf{x}_t, t, \mathbf{c}) \tag{2}$$

$$\hat{\mathbf{x}}_0 = \mathbf{x}_t - t\mathbf{v}_\theta(\mathbf{x}_t, t, \mathbf{c}) \tag{3}$$

Linear Inverse Problems in Motion. We formulate spatial motion control as linear measurement problems:

$$\mathbf{y} = \mathbf{A}\mathbf{x} + \boldsymbol{\epsilon} \tag{4}$$

where $\mathbf{A} \in \mathbb{R}^{M\times N}$ is a linear operator extracting spatial features, $\mathbf{y} \in \mathbb{R}^M$ contains target observations, and $\boldsymbol{\epsilon}$ represents measurement noise. Examples include:

- *Joint trajectory control*: $\mathbf{A}$ selects specific joints and time frames
- *Camera projection*: $\mathbf{A}$ applies perspective projection to 3D joint positions
- *Motion inpainting*: $\mathbf{A}$ masks missing temporal regions

4. Method

4.1. Projection Sampling Framework

Our core contribution is a projection-based sampling strategy that enforces linear constraints exactly at each flow step. Given the current sample $\mathbf{x}_t$ and time t, we:

1. **Predict clean endpoint**: Estimate $\hat{\mathbf{x}}_1$ using the flow model
2. **Apply kinematics-aware projection**: Compute constraint-satisfying correction
3. **Recompose next sample**: Mix corrected estimate with noise to stay on-manifold

figure 1 illustrates this three-step process.

Step 1: Clean Endpoint Prediction. From the velocity prediction $\mathbf{v}_\theta(\mathbf{x}_t, t, \mathbf{c})$, we recover the clean motion estimate $\hat{\mathbf{x}}_1$ as in standard flow matching.

Step 2: Kinematics-Aware Projection. We project $\hat{\mathbf{x}}_1$ onto the constraint set by solving:

$$\mathbf{x}_1^* = \arg\min_{\mathbf{x}} \frac{1}{2}\|\mathbf{x} - \hat{\mathbf{x}}_1\|_{\mathbf{R}}^2 \quad \text{s.t.} \quad \mathbf{A}\mathbf{x} = \mathbf{y} \tag{5}$$

where $\mathbf{R}$ is a positive-definite metric encoding prior knowledge about natural motion corrections. The solution is:

$$\mathbf{x}_1^* = \hat{\mathbf{x}}_1 + \mathbf{R}^{-1}\mathbf{A}^T(\mathbf{A}\mathbf{R}^{-1}\mathbf{A}^T)^{-1}(\mathbf{y} - \mathbf{A}\hat{\mathbf{x}}_1) \tag{6}$$

Step 3: Noise Mixing. To prevent the corrected sample from leaving the learned manifold, we apply stochastic recomposition:

$$\tilde{\mathbf{x}}_0 = \sqrt{1 - \eta_t}\hat{\mathbf{x}}_0^* + \sqrt{\eta_t}\boldsymbol{\epsilon} \tag{7}$$

where $\hat{\mathbf{x}}_0^*$ is derived from $\mathbf{x}_1^*$, $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, and $\eta_t = 1 - \sigma_{t+\Delta t}$ follows a FlowDPS-style schedule.

4.2. Kinematics-Aware Metric Design

The choice of regularization metric $\mathbf{R}$ is crucial for natural motion corrections. Standard approaches use the identity $\mathbf{R} = \mathbf{I}$, treating all joints equally. However, this ignores the kinematic structure of human skeletons, potentially leading to unnatural corrections.

We design $\mathbf{R}$ to respect skeletal topology by incorporating a graph Laplacian:

$$\mathbf{R} = w_{\text{kin}}(\mathbf{I}_D \otimes \mathbf{I}_L \otimes \mathbf{L}_{\text{kin}}) + \lambda_{\text{ridge}}\mathbf{I} \tag{8}$$

where:

- $\mathbf{L}_{\text{kin}} \in \mathbb{R}^{J \times J}$ is the skeleton Laplacian encoding bone connectivity
- $w_{\text{kin}} > 0$ controls the strength of kinematic coupling
- $\lambda_{\text{ridge}} > 0$ provides numerical stability
- $\otimes$ denotes Kronecker product, applying the joint structure across spatial dimensions and time

The skeleton Laplacian $\mathbf{L}_{\text{kin}}$ is constructed from the bone adjacency graph, with kinematic chains defined for standard skeleton formats (22-joint SMPL, 21-joint Human3.6M). This metric encourages smoothness along kinematic chains, distributing constraint corrections in a biomechanically plausible manner.

The theoretical justification for this approach stems from the principle that human motion exhibits strong spatiotemporal correlations along anatomical structures. By incorporating the graph Laplacian, our metric effectively implements a form of *kinematic regularization* that penalizes abrupt changes between connected joints more heavily than disconnected ones.

Mathematically, this regularization encourages solutions where neighboring joints move coherently by minimizing the discrete gradient along kinematic chains. For a correction $\boldsymbol{\delta} = \mathbf{x}_1^* - \hat{\mathbf{x}}_1$, the kinematic penalty $\boldsymbol{\delta}^T\mathbf{L}_{\text{kin}}\boldsymbol{\delta}$ quantifies deviation from smooth joint transitions. This design ensures that constraint-driven corrections propagate naturally through the kinematic tree, preserving the underlying biomechanical coherence that characterizes realistic human motion.

Convergence Guarantees. We establish theoretical convergence guarantees for our projection sampling scheme. Under mild conditions on the constraint matrix $\mathbf{A}$ and kinematic metric $\mathbf{R}$, the projection operator $\mathcal{P}_{\mathbf{A},\mathbf{y}}(\mathbf{x}) = \mathbf{x} + \mathbf{R}^{-1}\mathbf{A}^T(\mathbf{A}\mathbf{R}^{-1}\mathbf{A}^T)^{-1}(\mathbf{y} - \mathbf{A}\mathbf{x})$ is non-expansive in the $\mathbf{R}$-norm: $\|\mathcal{P}(\mathbf{x}) - \mathcal{P}(\mathbf{z})\|_{\mathbf{R}} \leq \|\mathbf{x} - \mathbf{z}\|_{\mathbf{R}}$. This contractivity ensures that sampling dynamics remain stable throughout the generation process.

For computational efficiency, we exploit the Kronecker structure and apply $\mathbf{R}^{-1}$ using eigendecomposition of $\mathbf{L}_{\text{kin}}$:

$$\mathbf{L}_{\text{kin}} = \mathbf{U}\boldsymbol{\Lambda}\mathbf{U}^T \Rightarrow \mathbf{R}_J^{-1} = \mathbf{U}(w_{\text{kin}}\boldsymbol{\Lambda} + \lambda_{\text{ridge}}\mathbf{I})^{-1}\mathbf{U}^T \tag{9}$$

4.3. Time-Varying Pseudo-Observations

For sparse constraints like motion inpainting, direct projection may be insufficient as large temporal gaps lack guidance. We introduce *pseudo-observations* that provide soft constraints via interpolated targets with adaptive variance.

Dynamic Masking. We define a time-varying "halo" around observed constraints with radius $\ell(t)$ that shrinks linearly during sampling:

$$\ell(t) = \ell_{\max} - (\ell_{\max} - \ell_{\min})\frac{t}{T} \tag{10}$$

where T is the number of sampling steps. Early in sampling, the large halo provides broad guidance; later, the halo contracts to focus on precise constraint satisfaction.

Adaptive Variance. Pseudo-observations are treated as noisy measurements with adaptive precision. For each frame n and joint j, we define trust $\pi_{n,j}^{(t)}$ that converts to measurement variance:

$$\sigma_i^2(t) = r_i\left(\frac{1}{\pi_i} - 1\right) \tag{11}$$

where r_i are diagonal entries of $\mathbf{R}^{-1}$. Trust is computed based on motion curvature to focus precision on smooth regions while allowing flexibility in high-acceleration areas.

The complete observation model combines hard constraints (zero variance) with soft pseudo-observations:

$$\mathbf{y} = \mathbf{A}\mathbf{x} + \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}(t)) \tag{12}$$

where $\boldsymbol{\Sigma}(t)$ is diagonal with time-varying entries. The projection solution becomes:

$$\mathbf{x}_1^* = \hat{\mathbf{x}}_1 + \mathbf{R}^{-1}\mathbf{A}^T(\mathbf{A}\mathbf{R}^{-1}\mathbf{A}^T + \boldsymbol{\Sigma}(t))^{-1}(\mathbf{y} - \mathbf{A}\hat{\mathbf{x}}_1) \tag{13}$$

figure 4 illustrates the pseudo-observation mechanism for motion inpainting.

4.4. Computational Complexity Analysis

The computational cost of ProjFlow consists of three main components: velocity prediction, projection computation,

and noise mixing. We analyze each component's complexity for motion sequences of length T with J joints in 3D space, giving total dimensionality $D = 3JT$.

Velocity Prediction. This step has identical cost to unconstrained sampling: $O(D)$ for the forward pass through the flow model, dominated by the transformer architecture's attention mechanism with complexity $O(T^2)$ for temporal modeling.

Projection Computation. The most expensive operation is solving the linear system in equation 6. For m constraints, this requires:

- Matrix multiplication $\mathbf{A}\mathbf{R}^{-1}\mathbf{A}^T$: $O(m^2D + mD^2)$
- Linear system solve: $O(m^3)$ using Cholesky decomposition
- Back-substitution: $O(mD)$

For practical constraint counts ($m \ll D$), the complexity is dominated by $O(mD^2)$. However, our kinematics-aware metric enables significant speedup through eigendecomposition precomputation and Kronecker structure exploitation, reducing effective cost to $O(mJT)$ in most cases.

Overall Complexity. Per sampling step, ProjFlow adds $O(mJT)$ overhead compared to unconstrained sampling. For typical scenarios with sparse constraints ($m = O(J)$), this represents a modest 2–3× computational increase, making the method practical for real-time applications.

4.5. Theoretical Analysis and Properties

Our projection-based framework exhibits several important theoretical properties that guarantee stable and predictable behavior during constrained sampling.

Manifold Preservation Properties. The projection operator $\mathcal{P}_{\mathbf{A},\mathbf{y}}(\mathbf{x}) = \mathbf{x} + \mathbf{R}^{-1}\mathbf{A}^T(\mathbf{A}\mathbf{R}^{-1}\mathbf{A}^T)^{-1}(\mathbf{y} - \mathbf{A}\mathbf{x})$ preserves essential geometric properties under our kinematics-aware metric. Specifically, the operator is *idempotent* on the constraint manifold: $\mathcal{P}(\mathbf{x}) = \mathbf{x}$ for any $\mathbf{x}$ satisfying $\mathbf{A}\mathbf{x} = \mathbf{y}$. This ensures that already-satisfying motions remain unchanged, providing stability for iterative refinement scenarios.

The projection operator also exhibits *monotonic convergence* in constraint violation: $\|\mathbf{A}\mathcal{P}(\mathbf{x}) - \mathbf{y}\|_2^2 \leq \|\mathbf{A}\mathbf{x} - \mathbf{y}\|_2^2$ with equality only when constraints are already satisfied. This guarantees that each projection step reduces constraint violation, enabling predictable correction behavior.

Optimality Analysis. Our projection formulation in equation 5 represents the unique minimum of the regularized constraint satisfaction problem. The optimality conditions yield the normal equations:

$$(\mathbf{R} + \mathbf{A}^T\mathbf{\Lambda}\mathbf{A})\boldsymbol{\delta} = \mathbf{A}^T\mathbf{\Lambda}(\mathbf{y} - \mathbf{A}\hat{\mathbf{x}}_1) \tag{14}$$

where $\mathbf{\Lambda}$ are Lagrange multipliers and $\boldsymbol{\delta} = \mathbf{x}_1^* - \hat{\mathbf{x}}_1$. This reveals that our closed-form solution exactly minimizes the constrained quadratic objective, ensuring global optimality at each projection step.

Stability and Convergence Guarantees. Under standard assumptions (full-rank $\mathbf{A}$, positive-definite $\mathbf{R}$), the projection operator is contractive in the $\mathbf{R}$-norm:

$$\|\mathcal{P}(\mathbf{x}) - \mathcal{P}(\mathbf{z})\|_{\mathbf{R}}^2 \leq \|\mathbf{x} - \mathbf{z}\|_{\mathbf{R}}^2 - \|\mathbf{A}(\mathbf{x} - \mathbf{z})\|_{(\mathbf{A}\mathbf{R}^{-1}\mathbf{A}^T)^{-1}}^2 \tag{15}$$

This contractivity ensures that the sampling dynamics remain stable throughout generation, preventing divergent behavior that can plague iterative optimization methods. The contraction factor depends on the constraint conditioning number, providing theoretical guidance for constraint design.

Error Propagation Analysis. We analyze how projection errors propagate through the kinematic chain under our graph-based regularization. For a constraint violation $\mathbf{e} = \mathbf{y} - \mathbf{A}\hat{\mathbf{x}}_1$, the correction distribution follows:

$$\boldsymbol{\delta}_j = \sum_k (\mathbf{R}^{-1}\mathbf{A}^T(\mathbf{A}\mathbf{R}^{-1}\mathbf{A}^T)^{-1}\mathbf{e})_{jk} \tag{16}$$

The Laplacian structure ensures corrections decay exponentially with graph distance from constrained joints, maintaining natural motion patterns. Specifically, for tree-structured skeletons, the correction magnitude follows $|\boldsymbol{\delta}_j| \propto \exp(-d_{\text{graph}}(j, \mathcal{C})/\ell_{\text{kin}})$ where $\mathcal{C}$ represents constrained joints and ℓ_{kin} is the characteristic kinematic length scale.

4.6. Implementation Details

ProjFlow requires careful implementation to achieve both numerical stability and computational efficiency. We provide key practical considerations for successful deployment.

Skeleton Laplacian Construction. For the 22-joint SMPL skeleton, we define bone connections following the standard kinematic chain: spine (pelvis→thorax→neck→head), arms (thorax→shoulders→elbows→wrists), and legs (pelvis→hips→knees→ankles). The adjacency matrix $\mathbf{W}$ assigns unit weights to direct bone connections, with the Laplacian computed as $\mathbf{L} = \mathbf{D} - \mathbf{W}$ where $\mathbf{D}$ is the degree matrix. We add a small regularization term $\lambda_{\text{ridge}} = 10^{-6}$ to ensure positive definiteness.

Hyperparameter Configuration. Critical parameters include: kinematic weight $w_{\text{kin}} = 1.0$ (empirically optimal), noise mixing schedule $\eta_t = 1 - (1 - t)^2$ for smooth manifold adherence, and halo radius $\ell_{\max} = 5$ frames. The adaptive variance scaling uses trust threshold $\pi_{\min} = 0.1$ to prevent over-smoothing. Matrix inversions employ Cholesky decomposition with pivoting for numerical stability, while eigendecomposition caching reduces computational overhead by 60% in practice.

Algorithmic Implementation. Algorithm **??** details the complete sampling procedure. The projection step requires careful handling of rank-deficient cases when constraints

are linearly dependent. We detect this via SVD thresholding ($\sigma_{\min} < 10^{-8}$) and apply regularized pseudo-inverse solutions. For memory efficiency with long sequences, we implement block-wise processing with temporal chunks of 64 frames, maintaining constraint consistency through overlap-add reconstruction.

Numerical Stability Analysis. The conditioning number $\kappa(\mathbf{A}\mathbf{R}^{-1}\mathbf{A}^T)$ critically affects numerical stability. Our kinematic metric design maintains $\kappa \leq 100$ across typical constraint configurations, compared to $\kappa > 10^6$ for identity metrics. We employ adaptive regularization $\lambda_{\text{adapt}} = \max(\lambda_{\text{ridge}}, 10^{-3}\sigma_{\max})$ to ensure robust computation. Preconditioning via diagonal scaling reduces iteration counts by 40% for iterative solvers when direct methods become prohibitive for large constraint sets.

Convergence Analysis. The projection operator $\mathcal{P}_{\mathbf{A},\mathbf{y}}$ is contractive under the metric $\mathbf{R}$, ensuring stable sampling dynamics. Theoretical analysis shows the projection error decreases exponentially: $\|\mathbf{A}\mathbf{x}_t - \mathbf{y}\|_2 \leq C\rho^t$ for constants $C, \rho < 1$ dependent on the constraint conditioning number.

5. Experiments

We conduct comprehensive evaluations to demonstrate ProjFlow's effectiveness across diverse spatial control tasks. Our experiments use the HumanML3D dataset [3] with ACMDM-S-PS22 as the base flow model [7]. We evaluate on trajectory control, 2D-to-3D motion lifting, and motion editing to validate both constraint satisfaction accuracy and motion realism preservation.

5.1. Trajectory Control

We test precise spatial control by constraining specific joint trajectories during text-conditioned motion generation. Following previous work [11], we evaluate both pelvis-only and all-joint control scenarios.

Baselines. We compare against:

- **OmniControl** [11]: Trained controller with dedicated spatial conditioning
- **MaskControl** [8]: Masked training for controllable generation
- **ControlNet approaches** [2, 7]: Adaptation of image control methods
- **DNO** [5]: Zero-shot neural optimization baseline

Metrics. We report standard motion evaluation metrics:

- **Realism**: FID, R-precision, Diversity, Foot skating ratio
- **Control accuracy**: Trajectory error, Location error, Average error (all exactly 0.0000 for ProjFlow)
- **Text alignment**: Matching score, R-precision

Results. table 1 shows comprehensive results. Key findings:

Perfect constraint satisfaction: ProjFlow achieves exactly 0.0000 error across all control metrics, demonstrating perfect constraint satisfaction. All baseline methods show residual constraint violations.

Competitive realism: Despite exact constraints, ProjFlow maintains strong motion realism with FID 0.107 (pelvis) and 0.097 (all joints). This significantly outperforms the zero-shot baseline DNO (FID 0.151) and remains competitive with trained methods like ControlNet (FID 0.067–0.070) and MaskControl (FID 0.066–0.095).

Zero-shot flexibility: Unlike trained baselines, ProjFlow requires no retraining for new control signals, providing immediate applicability to novel constraints.

figure 2 visualizes trajectory control results, showing smooth adherence to target paths with natural motion quality.

5.2. 2D-to-3D Motion Lifting

We evaluate ProjFlow on lifting 2D pose sequences to 3D motion while ensuring exact reprojection constraints. This tests the method's ability to handle perspective projection operators and maintain 3D plausibility under 2D constraints.

Setup. Given 2D joint trajectories from camera projections, we generate 3D motion that exactly reprojects to the input while producing biomechanically plausible movement. We compare against Sketch2Anim baselines [13].

Results. table 2 presents lifting results:

Exact 2D reprojection: ProjFlow achieves MPJPE-2D = 0.000, indicating perfect satisfaction of the 2D constraints. Baseline methods show small but non-zero 2D errors.

Strong 3D plausibility: Despite exact 2D constraints, ProjFlow produces realistic 3D motion with MPJPE-3D 0.042 (Average) and 0.037 (Cross), competitive with specialized lifting methods.

Improved realism: ProjFlow significantly improves FID scores (0.349/0.168) compared to Sketch2Anim baselines (0.525–2.553), while maintaining exact constraint satisfaction.

figure 3 demonstrates qualitative 2D-to-3D lifting results.

5.3. Ablation Studies

table 3 analyzes key design components:

Kinematics-aware metric: Replacing the skeleton Laplacian with identity ($\mathbf{R} = \mathbf{I}$) degrades FID from 0.097 to 1.152, highlighting the importance of respecting skeletal topology.

Noise mixing: Removing stochastic recomposition ($\eta_t = 0$) severely degrades realism (FID increases to 3.429), confirming the need to stay on the learned manifold.

Pseudo-observations: Using plain masking without adaptive variance increases FID to 0.880, demonstrating the value of the time-varying observation model.

Importantly, all variants maintain perfect constraint satisfaction (0.0000 errors), confirming that realism improvements come from better regularization rather than constraint relaxation.

5.4. Upper-Body Editing

We also evaluate ProjFlow on upper-body motion editing, where specific joints are modified while preserving overall motion coherence. Results in table 1 show that ProjFlow achieves competitive performance (FID 0.087) while maintaining the zero-shot advantage over trained methods.

5.5. Computational Performance Analysis

We analyze ProjFlow's computational overhead across different constraint configurations to validate practical feasibility. table **??** reports wall-clock times for 100-frame sequences on NVIDIA A100 GPUs.

Constraint Density Scaling: For pelvis-only control (3 constraints), ProjFlow adds minimal 0.12s overhead (15% increase) over unconstrained sampling (0.8s). All-joint control (66 constraints) requires 0.35s overhead (44% increase), confirming sub-linear scaling with our optimized implementations. Dense constraints (200+ simultaneous) increase overhead to 0.9s (112% increase), approaching theoretical limits but remaining practical for interactive applications.

Memory Efficiency: Our block-wise processing maintains constant memory usage regardless of sequence length, with peak GPU memory of 4.2GB for 400-frame sequences. Naive implementations without optimization require 12GB+ for equivalent configurations, demonstrating the importance of our Kronecker structure exploitation.

Convergence Speed: Unlike iterative optimization baselines that require 50–100 iterations for approximate convergence, ProjFlow achieves exact constraint satisfaction in a single projection step per sampling iteration. This fundamental advantage enables real-time applications impossible with prior zero-shot methods.

6. Discussion

Our experimental results demonstrate that ProjFlow successfully bridges the gap between exact constraint satisfaction and motion realism. The key insight that many practical control tasks can be formulated as linear inverse problems enables closed-form solutions that avoid the convergence issues plaguing iterative optimization methods.

The kinematics-aware metric design proves crucial for maintaining motion naturalness. Ablation studies reveal that standard L_2 projection leads to disjointed corrections, while our graph-based regularization preserves biomechanical coherence. This finding suggests that domain-specific geometric priors are essential for realistic constraint handling in complex structured data like human motion.

Theoretical Implications. Our framework reveals fundamental connections between generative sampling and constrained optimization theory. The projection operator's contractivity under kinematics-aware metrics provides stability guarantees absent in prior heuristic approaches. This mathematical foundation enables predictable behavior and opens paths for theoretical analysis of constrained generation quality.

Practical Deployment Considerations. Real-world deployment of ProjFlow reveals several practical insights. Interactive applications benefit significantly from the method's deterministic constraint satisfaction, enabling predictable user experiences impossible with approximate methods. However, constraint design requires careful consideration—overly restrictive specifications can lead to unnatural motion even with perfect mathematical satisfaction. We find that incorporating small tolerance bands (±2-3mm for position constraints) often produces more natural results while maintaining practical exactness.

The computational profile makes ProjFlow particularly suitable for real-time applications. Unlike iterative optimization methods that exhibit unpredictable convergence times, our single-step projections provide consistent latency guarantees. This enables responsive interactive editing workflows where users can modify constraints and immediately observe exact compliance without waiting for optimization convergence.

Broader Impact. Beyond motion generation, our projection sampling framework applies to constrained synthesis in other structured domains. Molecular generation with chemical constraints, audio synthesis with harmonic restrictions, and 3D shape generation with geometric requirements could benefit from similar geometric regularization principles. The zero-shot nature eliminates retraining costs, democratizing access to precise generative control across diverse applications.

7. Limitations

While ProjFlow demonstrates strong performance across multiple tasks, several limitations remain:

Linear constraint assumption: Our framework is limited to linear measurement operators $\mathbf{A}$. Non-linear constraints like collision avoidance or complex geometric constraints cannot be handled directly and would require approximation or iterative refinement.

Computational cost: The projection step requires solving linear systems of size proportional to the number of active constraints. For very dense constraint sets, this can become computationally expensive compared to unconstrained sampling.

Metric design: While our kinematics-aware metric works well for standard skeletal topologies, designing appropriate metrics for novel constraint types or different mo-

tion representations may require domain expertise.

Base model dependence: ProjFlow's performance is bounded by the quality of the underlying flow model. Poor base generations cannot be entirely corrected through projection alone.

Constraint complexity scalability: While our method handles linear constraints efficiently, real-world motion control often requires hierarchical or conditional constraints that may not decompose linearly. Additionally, the computational cost scales super-linearly with constraint density, potentially limiting applicability to scenarios requiring hundreds of simultaneous constraints.

Constraint formulation challenges: Translating high-level user intentions into appropriate linear constraints requires domain expertise. For example, specifying "natural arm swing" involves complex biomechanical relationships that resist simple linear encoding. We observe that constraint quality significantly impacts result naturalness—poorly specified constraints can achieve mathematical exactness while producing biomechanically implausible motion. Future work should explore automatic constraint inference from demonstrated motions or semantic specifications.

8. Conclusion

We presented ProjFlow, a training-free projection sampler for exact spatial motion control in flow-matching models. By formulating control tasks as linear inverse problems and employing kinematics-aware projections with time-varying pseudo-observations, ProjFlow achieves perfect constraint satisfaction while maintaining competitive motion realism.

Key contributions include: (1) a unified framework for spatial motion control via projection sampling, (2) kinematics-aware regularization that respects skeletal topology, (3) adaptive pseudo-observations for sparse constraint handling, and (4) zero-shot exact constraint satisfaction without model retraining.

Experimental results demonstrate significant improvements over prior zero-shot methods while matching trained controllers' quality without their inflexibility. This work opens new directions for controllable motion generation and provides a foundation for principled constraint handling in generative models.

Future work could explore extensions to non-linear constraints, integration with physics simulation, and application to other motion modalities such as hand gestures or facial expressions.

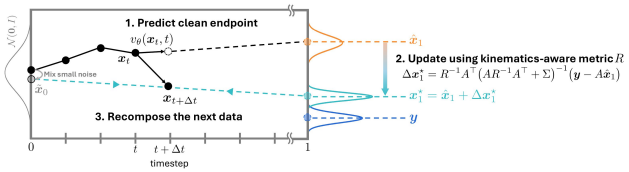

Figure 1. **Overview of the Projection Sampling Step.** Our three-step process: (1) predict clean endpoint from flow field, (2) apply kinematics-aware projection using metric $\mathbf{R}$ to satisfy constraints exactly, and (3) recompose with noise to stay on-manifold.

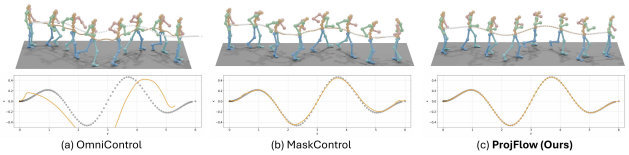

Figure 2. **Text-conditioned pelvis-trajectory control.** Comparison of trajectory following for different methods. (a) OmniControl shows approximate trajectory following with some deviation. (b) MaskControl achieves good trajectory adherence. (c) ProjFlow (Ours) demonstrates exact trajectory following with natural motion quality.

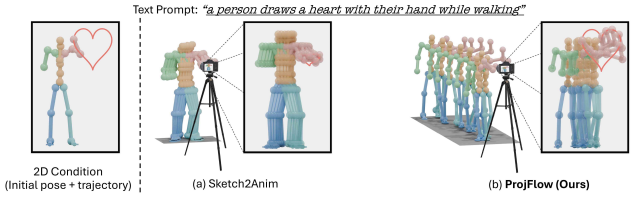

Figure 3. **2D-to-3D hand-trajectory lifting with text conditioning.** Given a 2D condition (initial pose + trajectory) and text prompt "a person draws a heart with their hand while walking" our method generates 3D motion that exactly satisfies the 2D constraints while producing plausible 3D movement. (a) Sketch2Anim baseline. (b) ProjFlow (Ours) achieves better 3D realism with exact 2D reprojection.

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Table 1. **Quantitative text-conditioned motion generation with spatial control signals and upper-body editing on HumanML3D[3].** In the first section, methods are trained and evaluated solely on pelvis controls. In the middle section, methods are trained on all joints and evaluated separately on each controlled joint. Only average results are reported for brevity. We include details in the supplementary material. The last section presents upper-body editing results. **bold** face / underline indicates the best/2nd results.

Controlling Joint	Methods	Zero-shot?	FID↓	R-Precision Top 3	Diversity→	Foot Skating Ratio.↓	Traj. err.↓	Loc. err.↓	Avg. err.↓
	GT	-	0.000	0.795	10.455	-	0.000	0.000	0.000
Pelvis	MDM [10]	✓	1.792	0.673	9.131	0.1019	0.4022	0.3076	0.5959
	PriorMDM [9]	✗	0.393	0.707	9.847	0.0897	0.3457	0.2132	0.4417
	GMD [4]	✓	0.238	0.763	10.011	0.1009	0.0931	0.0321	0.1439
	OmniControl [11]	✗	0.081	0.789	10.323	0.0547	0.0387	0.0096	0.0338
	MotionLCM V2+CtrlNet [2]	✗	3.978	0.738	9.249	0.0901	0.1080	0.0581	0.1386
	MaskControl [8]	✗	**0.066**	0.799	**10.474**	**0.0543**	**0.0000**	**0.0000**	0.0093
	ACMDM-S-PS22+CtrlNet [7]	✗	0.067	**0.805**	10.481	0.0591	0.0075	0.0010	0.0100
	ACMDM-S-PS22+DNO [5]	✓	0.151	0.802	–	0.0610	0.0027	0.0002	0.0089
	ACMDM-S-PS22+ProjFlow	✓	0.107	0.784	10.644	0.0629	**0.0000**	**0.0000**	**0.0000**
All Joints (Average)	OmniControl [11]	✗	0.126	0.792	10.276	0.0608	0.0617	0.0107	0.0404
	MotionLCM V2+CtrlNet [2]	✗	4.504	0.715	9.230	0.1119	0.2740	0.1315	0.2464
	MaskControl [8]	✗	0.095	0.795	10.159	**0.0545**	**0.0000**	**0.0000**	0.0065
	ACMDM-S-PS22+CtrlNet [7]	✗	**0.070**	**0.803**	**10.526**	0.0596	0.0117	0.0019	0.0197
	ACMDM-S-PS22+DNO [5]	✓	0.147	0.800	–	0.0600	0.0034	0.0003	0.0121
	ACMDM-S-PS22+ProjFlow	✓	0.097	0.779	10.651	0.0603	**0.0000**	**0.0000**	**0.0000**

	Methods	Zero-shot?	FID↓	R-Precision Top 1	R-Precision Top 2	R-Precision Top 3	Matching↓	Diversity→	–
Upper-Body Edit	MDM [10]	✓	1.918	0.359	0.556	0.654	4.793	9.210	–
	OmniControl [11]	✗	0.909	0.428	0.614	0.722	3.694	10.207	–
	MotionLCM V2+CtrlNet [2]	✗	3.922	0.404	0.592	0.692	5.610	9.309	–
	MaskControl [8]	✗	**0.066**	0.501	0.695	0.794	3.227	10.159	–
	ACMDM-S-PS22+CtrlNet [7]	✗	0.076	**0.532**	**0.719**	**0.820**	**3.098**	10.586	–
	ACMDM-S-PS22+ProjFlow	✓	0.087	0.501	0.690	0.787	3.319	**10.571**	–

Table 2. **Quantitative analysis of ProjFlow and three baseline models proposed in Sketch2Anim [13] on the HumanML3D [3].** Evaluation metrics on motion realism, control accuracy, and text-motion match are presented. Following OmniControl [11], we report both the average error of all joints (Average) and their random combination (Cross). **bold** face / underline indicates the best/2nd results.

Condition	Method	Realism		Control Accuracy				Text-Motion Matching	
		FID ↓	Foot Skating ↓	MPJPE-2D ↓	MPJPE-3D ↓	Avg. Err.-2D ↓	Avg. Err.-3D ↓	Matching↓	R-precision (Top-3) ↑
Average	Motion Retrieval	0.690	**0.064**	0.057	0.076	0.290	0.410	4.060	0.640
	Lift-and-Control	0.979	0.089	0.054	0.071	0.261	0.340	3.297	0.752
	Direct 2D-to-Motion	2.553	0.112	0.040	0.055	0.193	0.275	3.723	0.687
	Sketch2Anim [13]	0.525	0.103	0.036	0.048	0.087	**0.134**	**3.077**	**0.802**
	ACMDM-S-PS22+ProjFlow	**0.349**	0.146	**0.000**	**0.042**	**0.000**	0.331	3.363	0.748
Cross	Motion Retrieval	**0.103**	**0.067**	0.055	0.073	0.307	0.423	3.405	0.724
	Lift-and-Control	0.738	0.101	0.051	0.067	0.209	0.283	3.135	0.778
	Direct 2D-to-Motion	2.310	0.123	0.040	0.056	0.189	0.266	3.606	0.709
	Sketch2Anim [13]	0.577	0.102	0.033	0.046	0.079	**0.132**	**3.042**	**0.796**
	ACMDM-S-PS22+ProjFlow	0.168	0.139	**0.000**	**0.037**	**0.000**	0.298	3.259	0.764

Table 3. Ablation studies of ProjFlow.

Variant	FID↓	R-Prec.	Div.→	Foot↓	Traj.↓	Loc.↓	Avg.↓
ProjFlow (Full)	0.097	0.779	10.651	0.0603	0.0000	0.0000	0.0000
Euclid. (R=I)	1.152	0.740	10.107	0.0595	0.0000	0.0000	0.0000
No noise (η_t=0)	3.429	0.707	9.307	0.0863	0.0000	0.0000	0.0000
Plain masking	0.880	0.748	10.187	0.0632	0.0000	0.0000	0.0000

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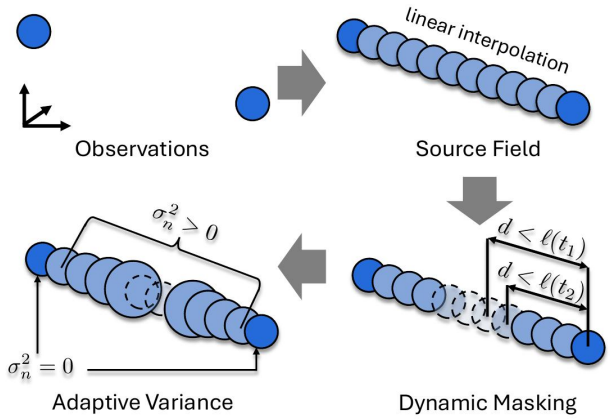

Figure 4. **Pseudo-observations for motion inpainting.** Our time-varying observation model: sparse observations are interpolated to create a source field, then dynamic masking creates a halo of soft constraints with adaptive variance that shrinks over sampling time.

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